John-Holt, Kingsley Bolobi

UNIVERSITY OF UYO

UNIVERSITY OF UYO
DEPARTMENT OF MATHEMATICS
2025/2026 FIRST SEMESTER EXAMINATION

COURSE:	MTH414: LEBESGUE MEASURE AND INTEGRATION
DATE SCHEDULED:	FRIDAY 6th MARCH, 2026
TIME ALLOWED:	TWO HOURS (2 HRS)

Instructions: Attempt any four (4) questions only.

1. (a) Define the term length $L(I)$ and find the length of the following: $17\frac{1}{2}$ *marks*
(i) $L((0,4])$ (ii) $L((7,14])$ (iii) $L((\infty,5])$.
(b) Let D be a dense set in $\mathbb{R}$. Let f be an extended real-valued function on $\mathbb{R}$ such that $\{x : f(x) > \alpha\}$ is measurable for each $\alpha \in D$. Show that f is measurable.
(c) Let f_1 and f_2 be nonnegative extended real-valued function on measurable set $D \subset \mathbb{R}$. Suppose $f_1 \leq f_2$ and f_1 integrable on D. Prove that $f_2 - f_1$ is defined a.e on D and $\int_D (f_2 - f_1) d\mu = \int_D f_2 d\mu - \int_D f_1 d\mu$.

2. (a) Define the following: (i) Additive set function (ii) Finitely additive set function $17\frac{1}{2}$ *marks*
(iii) Infinitely additive set function.
(b) State four properties of measure.
(c) State and prove he properties of additive set funcion over a ring.

3. (a) State the following terms: (i) Semiring of subsets of a set (ii) Ring of subsets of a set $17\frac{1}{2}$ *marks*
(b) State and prove the properties of a ring.
(c) Prove that a ring is a semiring.

4. (a) If $A(C)$ is the area of a circle, then $\mu(\mathcal{P}_*) \leq A(C) \leq \mu(\mathcal{P}^*)$ $17\frac{1}{2}$ *marks*
(b) Define the term Jordan measure.
(c) State and prove the properties of inner and outer Jordan measure.

5. (a) Define the following terms: (i) $\sigma-$additive (ii) Subadditive and (iii) superadditive of set functions. $17\frac{1}{2}$ *marks*
(b) State the Lebesgue outer measure.
(c) State the properties of Lebesgue outer measure.

6. (a) State without prove Lebesgue's Extension Theorem $17\frac{1}{2}$ *marks*
(iii) State the properties of Lebesgue integral of nonnegetive measurable function
(b) State without prove the Fatou's Lemma.